

*MA.7.GR.2.2***Benchmark**

MA.7.GR.2.2 Solve real-world problems involving surface area of right circular cylinders.

Benchmark Clarifications:

Clarification 1: Within this benchmark, the expectation is not to memorize the surface area formula for a right circular cylinder or to find radius as a missing dimension.

Clarification 2: Solutions may be represented in terms of pi (π) or approximately.

Connecting Benchmarks/Horizontal Alignment

- MA.7.NSO.1.1
- MA.7.AR.2.2
- MA.7.AR.4.4
- MA.7.GR.1.3, MA.7.GR.1.4
- MA.7.GR.2.3

Terms from the K-12 Glossary

- Cylinder (Circular)
- Pi (π)
- Surface Area

Vertical Alignment**Previous Benchmarks**

- MA.7.NSO.2.1, MA.7.NSO.2.2, MA.7.NSO.2.3
- MA.7.GR.1.2
- This is the first introduction to cylinders outside of recognizing the shape in early elementary.

Next Benchmarks

- MA.912.GR.4.6

Purpose and Instructional Strategies

In previous courses, students found the area of quadrilaterals and composite figures by decomposing them into triangles or rectangles, which developed into finding the surface area of right rectangular prisms and right rectangular pyramids using a figure's net. In grade 7 accelerated, students find the surface area of a right circular cylinder using the figure's net and build that into solving real-world problems involving surface area of right circular cylinders. In future courses, students will solve mathematical and real-world problems involving the surface area of cylinders, pyramids, prisms, cones and spheres.

- Instruction includes finding the height or the circumference when given the surface area of a right circular cylinder, but students will not be expected to find the radius as a missing dimension (*MTR.3.1*).

Common Misconceptions or Errors

- Students often confuse the vocabulary base, length, height and “B” (base area), when moving between two- and three-dimensional figures. To address this misconception, continue to use the parts of the net to calculate the surface area, rather than focusing on the formula.
- Students may incorrectly invert the height and base of a cylinder when the cylinder is presented horizontally. To address this misconception, remind students that while a cylinder may lay on its side, the bases are the parallel circles.
- Students may incorrectly apply the formulas for area, surface area and volume. To address this misconception, use these concepts in context, or a manner in which they understand the meaning behind the terms to foster their conceptual development (*MTR.7.1*).

Strategies to Support Tiered Instruction

- Instruction includes the use of geometric software to allow students to explore the difference between base, length, height and “B” (base area).
- Teacher creates and posts an anchor chart with visual representations of a right circular cylinder to assist in correct academic vocabulary when solving real-world problems.
- Teacher provides students with an example of a three-dimensional figure in its original position then provides multiple orientations to discuss how the location of the figure’s base changes, but the dimensions of the figure do not change.
 - For example, two right circular cylinders are shown below with the same dimensions but in different orientations. The base is highlighted in each.



- Teacher co-creates a visual of a three-dimensional figure and its dimensions in the context of a real-world problem.
- Instruction includes opportunities for students to solve for the surface area of a given right circular cylinder in terms of pi before replacing the value of pi with an approximation to determine the estimated surface area.
- Instruction includes co-creating a graphic organizer of a right circular cylinder and color-coding and labeling the dimensions.
- Teacher provides instruction focused on manipulatives or geometric software for students to develop understanding of the difference between the formulas for area, surface area and volume.
- Teacher provides opportunities for students to comprehend the context or situation by engaging in questions (laminating these questions on a printed card for students to utilize as a resource in and out of the classroom would be helpful).
 - What do you know from the problem?
 - What is the problem asking you to find?
 - Can you create a visual model to help you understand or see patterns in your problem?

- Teacher encourages students to continue to use the parts of the net to calculate the surface area, rather than focusing on the formula.
- Provide multiple orientations of cylinders, focusing on the height as the perpendicular distance between the parallel bases, continuing to break them down to their nets.

Instructional Tasks

Instructional Task 1 (MTR.4.1)

The Fine Arts Club will be making and selling a soda can snuggie for a fundraiser. They researched the dimensions of a standard soda can to be 4.83 inches high with a diameter of 2.13 inches across the top and 2.6 inches at the widest part of the can. A soda snuggie that will keep the soda cold will require an insulated layer, a liner and a decorative outer fabric.

Part A. Provide a design the Fine Arts Club could use to make their soda snuggie. How much of each material will be needed for each soda snuggie?



Part B. Compare your design with a partner (or group). What are the similarities? What changes (if any) would you make to your design based on the ideas of others?

Part C. Various companies have changed the size of cans for specialty drinks such as coffee and flavored water drinks. Research other size cans and compare the surface area of the various cans to the soda can. Determine how much material would be needed for each. Compare potential costs for the Fine Arts Club if they wanted to offer any of these additional sized snuggie options.

Instructional Items

Instructional Item 1

A cosmetics company is selling a new line of lipstick and needs to determine how much plastic is needed to wrap each cylindrical tube. If the lipstick tube is 12.1 millimeters (mm) in diameter with a length of 72 mm, how many mm^2 of plastic is needed for one tube? How much will be needed for a box of 24?

Instructional Item 2

Melanie is buying a candle for a gift. She has 80.07 in^2 of wrapping paper and all of the candles she is looking at have a radius of 1.5 in. What height candle can Melanie buy if she uses all of the wrapping paper she has?

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*